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4D Printing of Continuous Shape Representation

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4D printing can address time-evolving structural functions that are unattainable by conventional 3D printing. Despite the advance in materials and printing techniques, however, 4D printing of continuity of shape representation that generally characterizes 3D matters is still challenging, because the existing methodologies mostly rely on a few discrete levels of strain and their spatial distributions. Here, a 4D printing strategy of shape memory polymers (SMPs) that can program continuous levels of shape-recovery strain is proposed. It is found that the irrecoverable state of the SMP and the corresponding recovery strain can be controlled in a continuous and precise manner by a single printing parameter. Importantly, the continuity of strain programming provides an opportunity for the translation into mathematical function representation (F-rep), which allows the systematic derivation and implementation of 4D-printed bilayer strain functions that are matched to the continuously varying curvatures of the target geometry. Combined with the custom-built software, the F-rep 4D printing strategy can produce 4D-printed architectures that involve continuously varying strain profiles of almost any function type. The effectiveness of the framework is highlighted by a set of 3D face masks with facial feature transformation driven by a function operator.

1. Introduction

4D printing is an emerging technology that provides a prototypical route for self-shaping growth, assembly, and actuation of 3D matters.[1,2] With a surge of shape-morphing active materials and advanced additive manufacturing technologies, this newly developed additive manufacturing strategy allows for time-evolving, targeted shape changes of 3D-printed structures by patterning shape-shifting active materials that respond to external stimuli, such as heat,[3–8] humidity,[2,9] solvents,[10,11] and magnetic fields.[12] Central to the 4D printing technology is the spatial control of the "extradimensional" mechanical responses (i.e., strain) of 3D-printed structures. Typical examples include the spatial patterning of the shape-shifting behavior of shape memory polymers (SMPs),[3–7] swelling ratio of hydrogels,[2,9] nematic order of liquid crystal elastomers,[8] and magnetic domains.[12] A common modality of these responses stems from the internal strain distributions programmed into the 3D-printed active materials, which form target local curvatures of 3D structures. Since a 3D structure is generally defined by a complex set of curvatures, 4D printing ideally requires the programming of continuously varying strains with values matched to each infinitesimal local curvature. However, in numerous cases, fundamental challenges arise at the point of translating the continuity of such shape representation of 3D matters into only a few discrete levels of strain and their spatial distributions. Indeed, the majority of the existing approaches have relied on manual stretching[3,13,14] or spatial patterning of a few strain levels in combination with passive layers.[5,15–17] Although pioneering studies have addressed such issues by using sophisticated origami hinge designs,[6,7] modified molecular filler orientations,[8,18] and multimaterial printing techniques,[19] the facile programming of continuous strain profiles is still promising for broad potential applications of 4D printing.

Here, we propose a new 4D printing strategy of SMPs that can program continuous levels of shape-recovery strain (ε_r). In our printing strategy, the thermomechanical training and irrecoverable state of the SMP can be engineered in a continuous manner by only a single printing parameter, the extrusion feed rate. The result is the direct and continuous ε_r programming along with arbitrary printing paths. To maximize the effectiveness of such continuity of strain programming, we introduced the mathematical function representation (F-rep) of shape, strain, and printing parameters that provided a systematic and interpretable framework for 4D printing. The resulting 4D-printed architectures could involve continuously varying ε_r profiles of arbitrary function forms, including polynomial,

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exponential, trigonometric, and their composites, and their time-evolving transformations exhibited unprecedented design versatility and accuracy. The generalization of this approach was achieved by the functionally represented discretization of arbitrary complex 3D surfaces with the help of rationally designed supportive architectures. As an example, we demonstrated a set of 3D face masks, which can be defined from a 3D model by mathematical functions, modified their facial features by multiple function operators, and reproduced them in 4D-printed architectures.

2. Results and Discussion

2.1. F-Rep of Continuous Strain Programming

An arbitrary 3D matter generally possesses a complex set of (or, in many cases, continuously varying) curvatures; thus, the conventional approaches of discrete strain programming necessarily carry a non-negligible degree of shape approximation (**Figure 1**a–c). By contrast, our approach begins with the mathematical F-rep of the target complex structure (Figure 1d). Theoretically, F-rep can translate the surface of an arbitrary complex 3D geometry into an implicit function form, $f(x, y, z) = 0$.[20,21] For simplicity, in this study, we focus on the quasi-3D geometry that has a uniform value for an infinitesimal distance (δy) along the y-axis (see Figure 1a,d). Considering that any given 3D geometry can be redefined by a set of 2D slices which are cut in arbitrary directions, this geometric simplification does not contradict the generalization process of our strategy mentioned below. Given this definition, the F-rep of the target structure is reduced to the following shape function (f) and the corresponding curvature function (κ) (Figure 1e)

$$f(x, z) = 0 \tag{1}$$

$$\kappa = \frac{f_x^2 f_{xx} - 2 f_x f_z f_{xz} + f_x^2 f_{zz}}{\left(f_x^2 + f_z^2\right)^{3/2}} \tag{2}$$

where $f_{i,x}$, $f_{i,z}$, $f_{i,xx}$, $f_{i,xz}$, and $f_{i,zz}$ are partial derivatives. To translate such shape information into the target, continuously varying strain profiles for 4D printing, we combined the above definition with the Timoshenko equation which suggests that a locally defined curvature can also be interpreted as a function of strain mismatch in bilayer constructs[22,23]

$$\kappa = \frac{6mn(1+m)}{1+4mn+6m^2n+4m^3n+m^4n^2}\frac{\varepsilon_t - \varepsilon_b}{h_b} \tag{3}$$

where m is the ratio of the layer thicknesses, $m = h_t/h_b$, n is the ratio of the Young's moduli, $n = E_t/E_b$, and ε_t and ε_b are strain

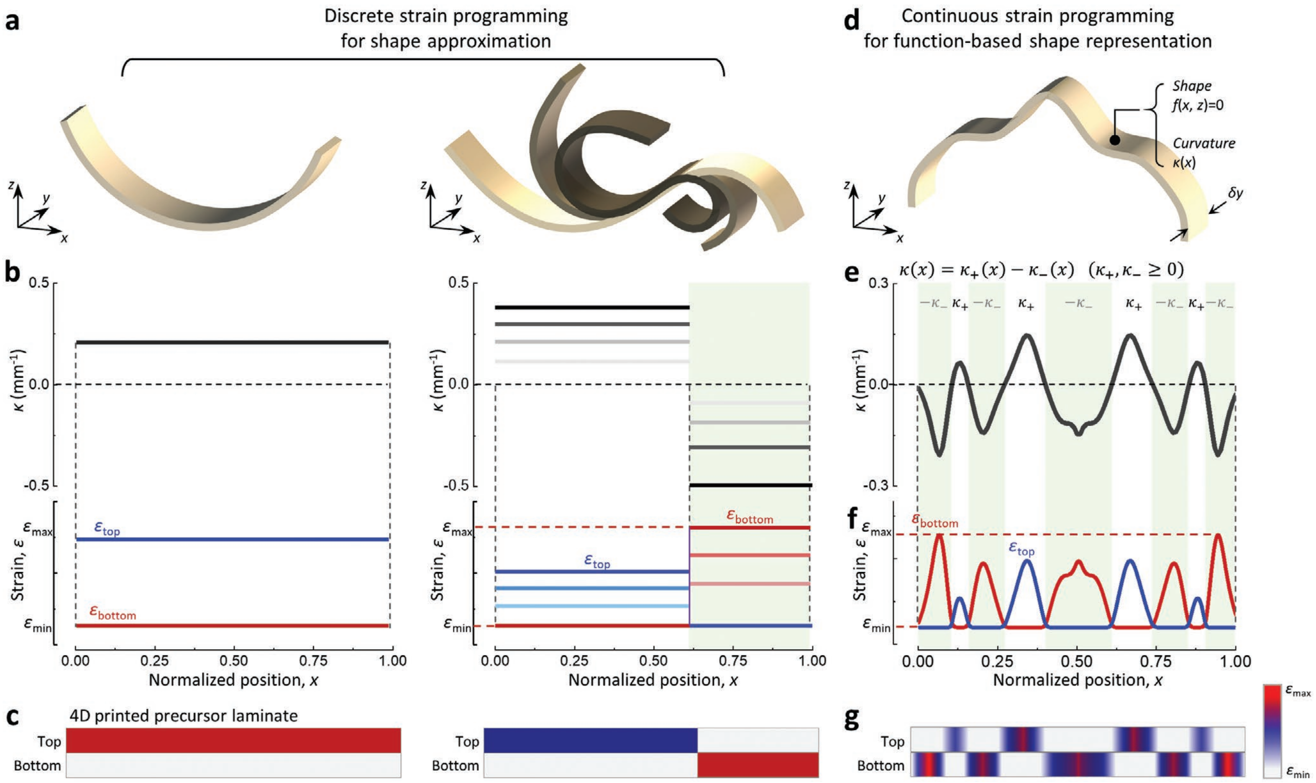

Figure 1. The concept of the F-rep of continuous strain programming. a) Schematic illustrations of typical 4D-printed matters with discrete strain programming. b) Continuous curvature profiles and the corresponding discrete bilayer strain profiles for shape approximation in (a). c) Cross-sectional schematic of 4D-printed precursor laminates with the bilayer strain profiles shown in (b). d) Schematic illustration of a complex shape-shifting matter with continuously varying curvatures and strains. e) Continuous curvature profiles of (d) and the function-based coordination. f) Function-based coordination of continuous bilayer strain profiles for interpretable shape representation. g) Cross-sectional schematic of the 4D-printed precursor laminate with the bilayer strain profile shown in (f).

functions, respectively, with subscripts t and b, which denote the top and bottom layers. Here, we postulated that continuously varying curvatures programmed in bilayer constructs can be generally represented by their mathematical function form $\kappa(x)$ from Equation (3)

$$\kappa(x) = \frac{c}{h_b}\{\varepsilon_t(x) - \varepsilon_b(x)\} \tag{4}$$

where c is constant assuming that the thickness and material properties are the same for the top and bottom layers. By exploiting a simple mathematical trick regarding absolute value functions, on the other hand, we can obtain another structurally equivalent form of $\kappa(x)$ (Figure 1e)

$$\kappa(x) = \kappa_+(x) - \kappa_-(x) \tag{5}$$

where $\kappa_+(x)$ and $\kappa_-(x)$ are defined in piecewise fashion by $|\kappa(x)|$ for x satisfying $\kappa(x) \geq 0$ and $\kappa(x) < 0$, respectively, and 0 for others. The resulting relation implies that $\kappa(x)$ can be directly divided into two positive functions, ε_t and ε_b, as $\varepsilon_t = \kappa_+/c + \varepsilon_{min}$ and $\varepsilon_b = \kappa_-/c + \varepsilon_{min}$ based on the feasible range of ε_r programming that is necessarily positive according to SMP properties and the 4D printing setup, where ε_{min} is the minimum recovery strain that can be programmed (Figure 1f). To complete the function relation, it is necessary to rescale the domain of interest, particularly $\{x\}$ as shown in Figure 1d, due to the change in total length of the SMP laminates under actuation. Based on the minimal theoretical framework combined with aspects of the Timoshenko equation, we introduced a new domain $\{s\}$ that satisfies the following equation (Figure S1, Supporting Information; see the Experimental Section for details)

$$\mathrm{d}s = \frac{\sqrt{1+\left(f_{i,x}\right)^2}}{1 - \frac{2h}{3}|\kappa| - \varepsilon_{min}}\,\mathrm{d}x \tag{6}$$

The results are significant in that the target functionally represented shape is directly related to the functionally represented continuous strain profile of its 4D-printed precursor (Figure 1g).

2.2. Principle of Continuous Strain Programming in SMP 4D Printing

To program the derived ε_r functions above, we devised a one-step, single-material 4D printing strategy of SMPs that enables direct and continuous ε_r programming along 3D-printed unit SMP fibers without additional pre- and postprocesses (**Figure 2**). In general, the shape-shifting property of a SMP is determined by the existence of two different molecular configurations, a permanent shape and a temporary shape, which can be controlled by a thermomechanical training process that involves heating (above the glass transition temperature T_g) and mechanical loading steps.[1,24] We elaborately integrated this thermomechanical training procedure of SMPs into the process of FDM 3D printing to implement effective strain-printing parameter translations (Figure 2a–c and Figure S2 and Movie S1 (Supporting Information)). We pointed out that the extrusion and deposition processes of FDM printing intrinsically carry a sequence of heating and mechanical loading steps, both of which are key elements for the thermomechanical training of SMPs.[6,25] Specifically, the SMP filament (diameter D) is first heated at a nozzle up to the printing temperature T_p ($>>T_g$) and then extruded in the form of a thin fiber (diameter d) at an extrusion feed rate of C (Figure 2b(i–iii)). During this process, the filament is stretched by the geometric constraint defined by the cross-sectional area ratio of the nozzle and filament, $\lambda_e = D^2/d^2$. The extruded fiber is then deposited and swiped on a bed, on which the fiber and internal polymer chains experience additional mechanical stretching (λ_d) depending on the cross-sectional area ratio between the as-extruded and the as-deposited fibers; namely, $\lambda_d = \pi d^2/4wh$, where w and h are the width and height of the deposited geometry, respectively (Figure 2b(iv) and Figures S2 and S3 (Supporting Information) and the Experimental Section). Our key observation here is that the total mechanical stretching ($\lambda_{total} = \lambda_e\lambda_d$) involved in the printing process cannot fully deliver the calculated ε_r upon exposure to an extremely harsh thermomechanical condition ($T_p >> T_g \approx 55$ °C) that occurs during the extrusion process (Figure 2c and Figure S4 (Supporting Information)). The occurrence of this irrecoverable regime is mainly attributed to the amount of heat transfer to SMPs, which can be changed by controlling the temperature and exposure time (t_h).[26] We also found that the programming of the irrecoverable regime, albeit indirectly, led to elaborate ε_r programming. In our printing strategy, such heat transfer could be accurately and continuously controlled by changing the extrusion feed rate C ($\sim t_h^{-1}$). Therefore, the control of C provided a facile solution to achieve continuous ε_r programming in spatial domains by engineering the SMP irrecoverable rate (Figure S4, Supporting Information).

2.3. Implementation and Characterization of Continuous Strain Programming

Our strategy of continuous ε_r programming in SMP 4D printing is highly dependent on the way of C control. Therefore, much effort was invested to investigate the quantitative relationship between ε_r and C. We first established a strategy for independent C control using the relation between C and the printing speed F, i.e., $C = F/\lambda_d$ (see the Experimental Section for details). Based on this control scheme, we fabricated various 4D-printed linear specimens with varying C, and analyzed their shape memory effects (Figure S5, Supporting Information). The pure effect of C on ε_r was investigated by comparing the shape memory effects of the following two types of printed specimen: i) one was 4D-printed without the deposition process and then immediately cooled down at room temperature, and ii) the other was 4D-printed with deposition process but without additional mechanical stress by setting $\lambda_d = 1$ (Figure 2d). The results show that both samples yielded the same ε_r responses, indicating that C can be uniquely related to ε_r independent of the deposition process. This ε_r–C relationship is not only unique for the constant parameter set of (T_p, λ_d), but is also found to have a one-to-one correspondence characterized by the following exponential decay form with R^2_{adj} ranging from 0.969 to 0.997

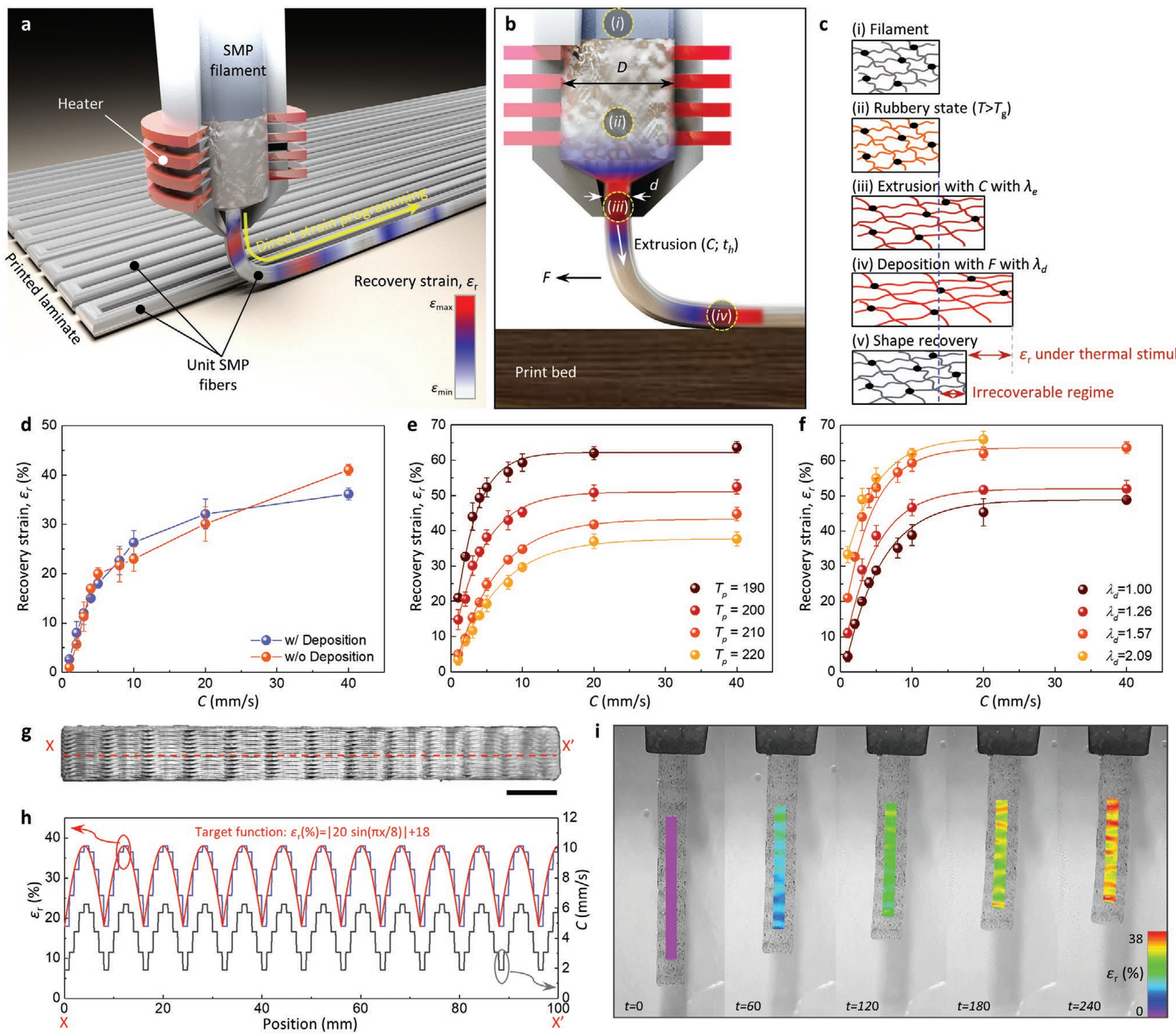

Figure 2. SMP 4D printing for continuous strain programming. a) Schematic of the SMP 4D printing process that enables direct and continuous ε_r programming in a unit fiber level. b,c) Key printing parameters (b) and the integrated thermomechanical training mechanism with an irrecoverable regime (c). d) Measured ε_r responses as a function of C. The results with and without the deposition process are both in good agreement, which indicates that the ε_r–C relation is independent of the deposition process. e,f) Measured ε_r responses with various T_p (e) and λ_d (f). Each line represents the corresponding characteristic ε_r–C curves. g) Photograph of a 4D-printed precursor laminate for strain analysis. Scale bar, 1 cm. h) Programmed strain function and its translation into C based on the characteristic ε_r–C curve. i) Sequential image frames of the sample in (g) under heat-induced shape transformation. Digital image correlation is used to visualize the surface strain field which is predicted from the programmed ε_r profile in (h).

$$\varepsilon_r = \varepsilon_{sat}\left(1 - e^{-\frac{C}{\alpha}}\right) \tag{7}$$

where ε_{sat} and α denote the saturated maximum recovery strain and decay constant, respectively. Additional experimental results obtained with various λ_d and T_p show the programmable range of ε_r (roughly 0.05–0.65) and the corresponding characteristic ε_r–C curves (Figure 2e,f). Hereafter, we adopted the parameter set of (T_p = 200 °C, λ_d = 1.57) with C varying from 1 to 40 mm s^{-1} as the standard setting condition with the ε_r–C F-rep relation of $\varepsilon_r = 0.4844[1 - \exp(-C/0.2455)]$, which provides stable printing within a relatively wide programmable strain range of $0.15 \le \varepsilon_r \le 0.48$.

To enable the continuous translation of ε_r into C based on the characteristic ε_r–C curves, we developed a custom-built 4D printing software (Figure S6, Supporting Information). The software was designed to split up printing paths into a set of equidistant unit segments and assign adequate C values to them by modifying each G-code. Therefore, length of the unit segment (dl) indicates the programmable resolution of ε_r, which was normally set to 1 mm. Note that there exist, in principle, up to 2340 levels of ε_r that can be programmed into each dl, since the ε_r–C F-rep relation allows precise control of ε_r by changing C values with the resolution of 1 mm min^{-1} (see the Experimental Section for details on the software). Importantly, the software also provided the user interface to program any type of F-rep of ε_r into the corresponding C through the user-defined ε_r–C relation as described in Equation (7). This function allowed us to precisely design a 4D-printed structure where continuously varying ε_r, which was translated from the functionally represented shape information as explained in the Section 3.1, was programmed; here, we call this framework as "F-rep 4D printing" which implies the entire sequence of i) the F-rep of shape and strain, ii) the physical translation of strain function (continuous profile) into the printing parameter, and iii) the printing process. The proposed F-rep 4D printing was experimentally verified using surface strain analysis. A 4D-printed sample programmed with a periodic ε_r function through the 1 mm resolution C control was prepared and analyzed (Figure 2g,h). Direct visualization of surface strain fields suggests that complex ε_r profiles were continuously programmed over relatively large spatial domains (≈10 cm) and that the resulting periodicity and the maximum value were matched to the expected ones (Figure 2i; see the Experimental Section for details on surface strain mapping).

2.4. Demonstrations of Functionally Represented Shape-Shifting Matters

To validate the effectiveness of our F-rep 4D printing framework, we demonstrated multiple F-rep 4D-printed slices (**Figure 3**). Each slice was programmed based on the 2D target shapes shown in Figure 3a, whose curvature functions were selected as the form of arithmetic operations and/or composition of polynomials, trigonometric, and exponential functions, indicating that our approach can be readily extended to arbitrary shapes. By leveraging the function-coordination scheme based on the shape-strain relation as shown in Figure 1e,f, we successfully translated the curvature functions into the bilayer strain functions (ε_t and ε_b) (Figure 3b). Given the programmable strain range ($0.15 \le \varepsilon_r \le 0.48$) for the standard parameter set (T_p =200 °C, λ_d = 1.57), we rationally set ε_{min} to 0.18 with the programming resolution of 1 mm, which ensured stable 4D printing (Figure 3c). Each 4D-printed precursor slice consisted of four layers in total: two for the top layers (thickness of 0.4 mm) and the other two for the bottom layers (thickness of 0.4 mm). Under uniform heat (90 °C), all samples were transformed into programmed shapes within 2 min (Figure 3d and Movie S2 (Supporting Information)). The resulting shapes were in good agreement with the targeted ones as shown in Figure 3a, and their curvature profiles validated the accuracy of our approach, illustrating the enhanced programmability of our design strategy in an interpretable and reproducible manner (Figure 3e).

2.5. F-Rep 4D Printing and Function-Based Modification of Complex 3D Surfaces

The creation of complex, 3D surfaces from lower-dimensional flat sheets is of increasing interest in fields ranging from origami,[27,28] kirigami,[29–31] conformable electronics,[32–34] and soft robotics.[35,36] By extending our F-rep 4D printing method, herein, we propose a 4D printing framework that can not only rationally reconstruct arbitrary 3D surfaces, but also systematically modify the original design. The basic approach is twofold: 1) cutting the target 3D surface model into multiple slices and implementing the F-rep 4D printing for each of them, and 2) connecting each slice with sets of tiny, thin interconnecting lines to create a smooth surface. Specifically, our strategy reconstructs complex 3D structures with multiple values of Gaussian curvature via combination of the engineering of bending stiffness of the shape-shifting frames and interconnecting layers. In this approach, each slice plays the role of the core backbone that causes dominant deformations (shape-memory effect) under stimuli, whereas the interconnecting lines are engineered to naturally follow the shape morphing of the backbone structures (see the Experimental Section).

As an illustrative example, we demonstrated 4D-printed face masks whose design was simply modified from one of the most popular Korean traditional masks (**Figure 4**a). As explained above, we first remodeled the target face mask design into an array of 19 slices which served as core backbone frames of the shape-shifting surface (Figure 4b, left and Figure S7 (Supporting information)). Then, the shape of each slice was functionally represented by curve fitting with the 8th-order Fourier series expansion in explicit forms (Figure S9 and Table S1, Supporting Information). Due to the slicing process, the fidelity of the 4D reconstruction along the y-axis increased with the number of the slices, while the maximum programmable resolution of the precursor laminate which governs x- and z-axis shape was 0.4 mm. To successfully reconstruct the smooth 3D surface in the form of interconnected 2D slices, we additionally designed three types of supportive architectures: 1) interconnecting lines, 2) boundary structures, and 3) hinges (Figure S7, Supporting Information). First, the interconnecting lines were employed to passively form interpolative connections between adjacent slices so as to create a loft between two profiles in 3D modeling. The bending stiffness of the interconnecting lines with a width and a height of 0.4 mm was engineered to be much lower (0.46%) than that of the core slices against deformation in z-axis. Therefore, the interconnecting lines could form interpolative bridges between adjacent slices in an approximate manner even under their shape transformations. In addition to the interconnecting lines, boundary structures with thickness of 2.4 mm were added to both ends of the slices as geometric boundary conditions which elaborately coordinate relative movements of each slice under transformations without any postprocess. These structures were programmed to linearly shrink with strain rates distinct in every slice, in order to maintain the overall rectangular frame both in the as-printed and the as-actuated states. Moreover, the increased stiffness, due to the large thickness, allowed them to serve as reinforcements which compensate for undesirable loading effects (i.e., sagging) during transformation. Finally, hinges were necessarily programmed to induce out-of-plane transformation of the slices from the initially flat precursor laminate geometrically constrained by boundary structures. The desirable angle of the hinge could be approximated with a slope of the tangent of the target 2D curve at the edge. Since the theoretical

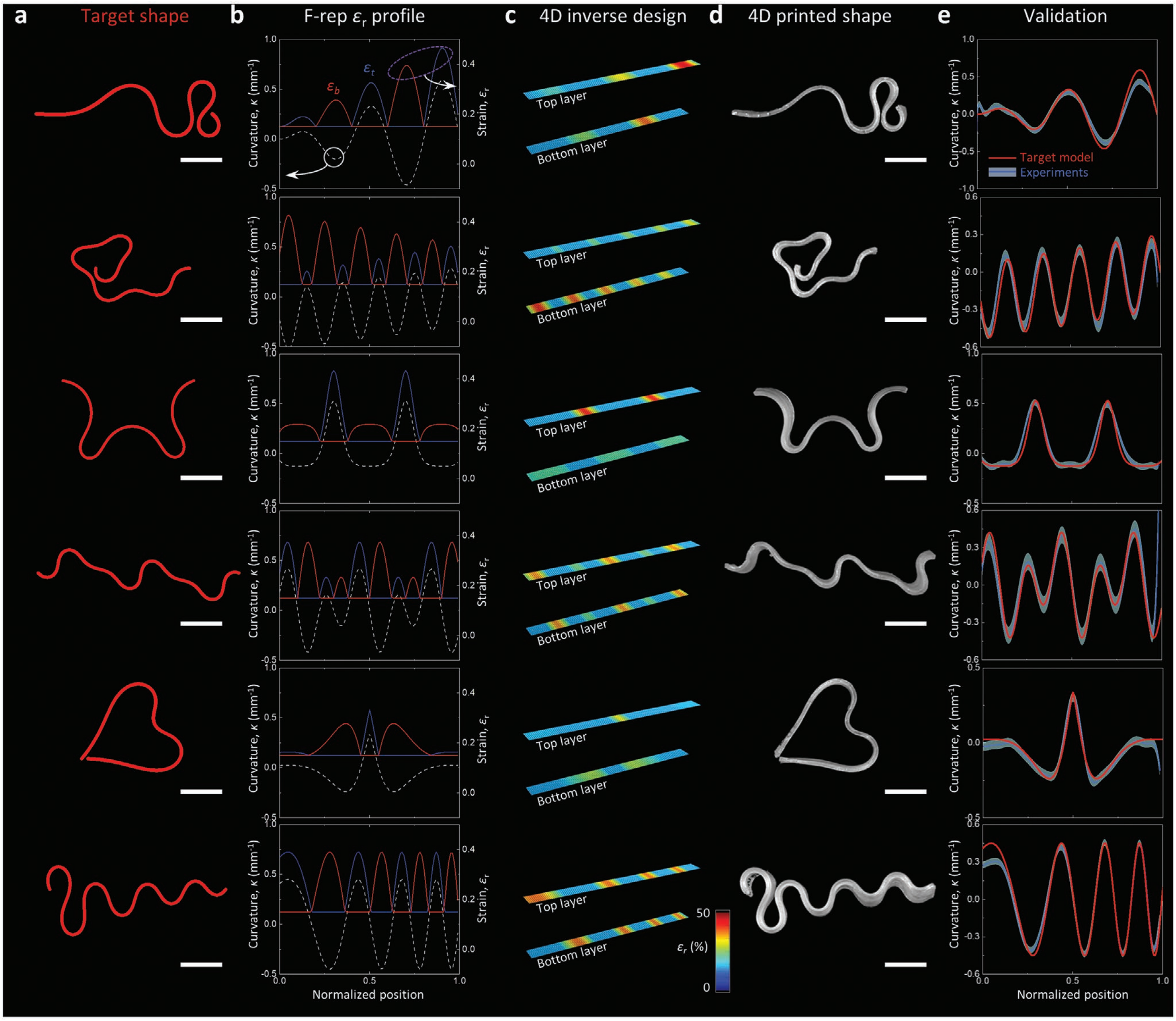

Figure 3. F-rep 4D printing of complex quasi-3D matters. a) Target shape models whose curvatures are defined by arithmetic operations and/or composition of polynomials, trigonometric, and exponential functions. Curvature functions are defined as follows in order from the top, i) $\kappa(\bar{x}) = 0.656\bar{x}\sin(5\pi\bar{x})$, ii) $\kappa(\bar{x}) = 0.234(\bar{x}-1) - 0.3\sin(10\pi\bar{x})$, iii) $\kappa(\bar{x}) = 0.06 \times e^{-2.4\sin(5\pi\bar{x})} - 0.131$, (iv) $\kappa(\bar{x}) = 0.188\cos(5\pi\bar{x}) + 0.281\sin(10\pi\bar{x})$, v) $\kappa(\bar{x}) = \dfrac{1181.25(\bar{x}-0.16)(\bar{x}-0.45)(\bar{x}-0.55)(\bar{x}-0.84)}{e^{15|\bar{x}-0.5|}}$, vi) $\kappa(\bar{x}) = 0.45\sin(5\pi(\bar{x}+0.27)^2)$, where $\bar{x}$ denotes the normalized position along the printing path. b) Shape-strain translation. c) Bilayer strain distributions programmed for 4D printing. d) Photographs of 4D-printed laminates after transformation. e) Validation of curvature profiles in comparison with the target model in (a). All scale bars are 1 cm.

maximum curvature that can be achieved with the given programmable strain range ($0.15 \le \varepsilon_r \le 0.48$) and thickness of the hinge (0.8 mm) is 0.62 (see Equations (3) and (4)), it was able to induce the out-of-plane motion of the mask by generating steep curvatures at the edge of the mask.

Distinct from other existing approaches, the most attractive feature of our strategy is the ability to mathematically modify the target model by applying appropriate function operators.[20,21] For verification, we introduced a customized weight function, $W(x, y)$, which was defined to be 1 for slices except those located in the part of the nose and cheek (specifically, 6th–10th slices) (Figure 4b, inset). The nature of the weight function permitted function modification through a simple calculation, i.e., $g = f \times W$, where f is the F-rep of the initial design and g is that of the modified design; the result predicted both additional protrusion of the nose and the retraction of the cheek (Figure 4b, right). The designed 3D models were translated into ε_r profiles using the function-coordination scheme shown in Figure 1e,f (Figure 3c). The resulting 4D-printed precursor laminates had overall dimensions of 16.6 cm × 14 cm × 2.4 mm including additional supportive structures (Figure 4d and Figure S8 (Supporting Information)). Shape transformation was performed for 5 min by immersing the precursor laminate, which was attached to buoys to compensate for the

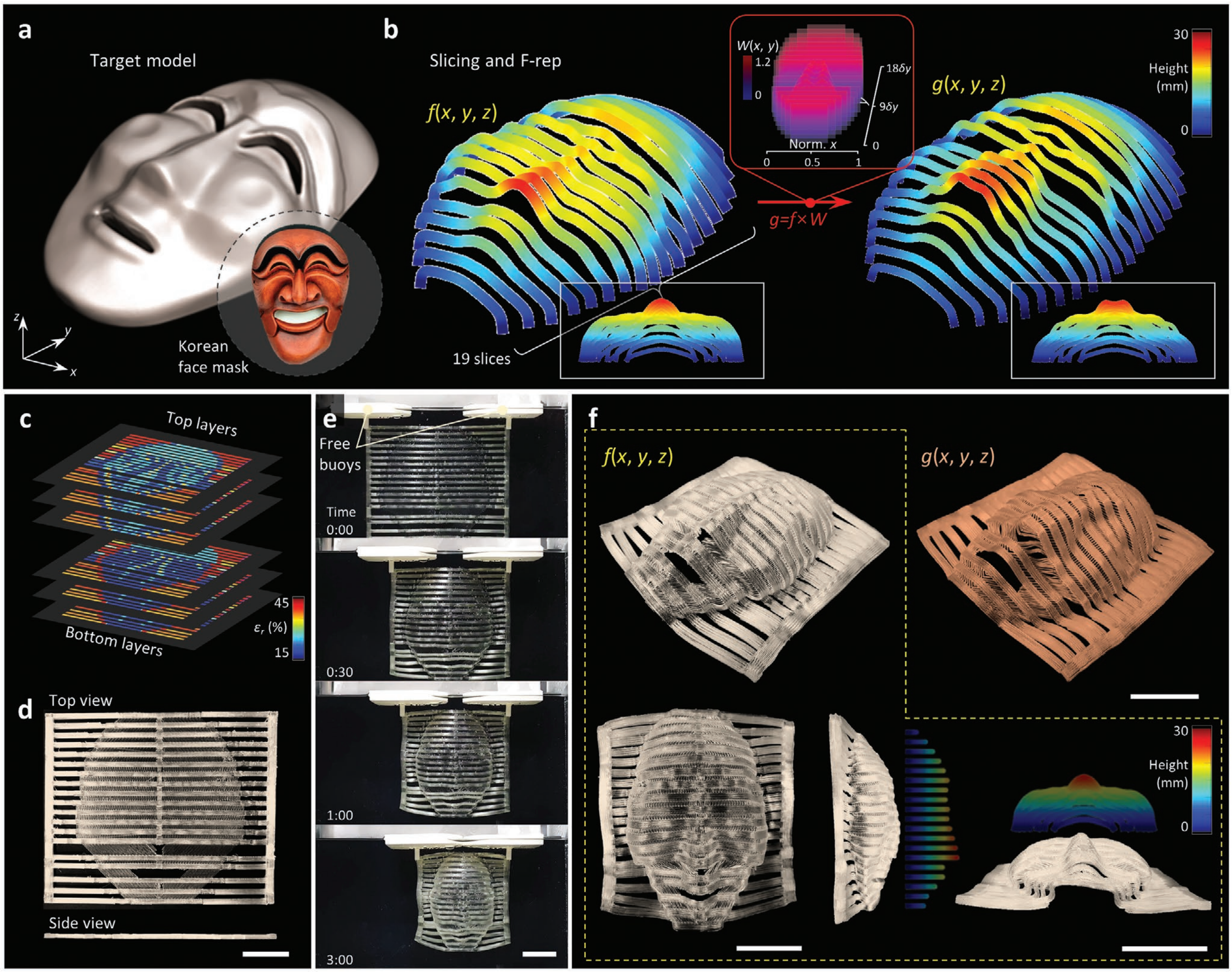

Figure 4. F-rep 4D printing and function-based modification of complex 3D surfaces. a) A 3D model of a target face mask, the design of which is reduced from one of the most popular Korean face masks shown in the inset image. b) Multiple slicing (19 slices) and F-rep of the target model ($f = 0$) and design modification ($g = 0$) through the function operation which is defined as a customized weight function, $W(x, y)$. Inset image highlighted by red contours: the exact profile of $W(x, y)$. c) Strain distributions of top and bottom layers for 4D printing programmed by a custom-built software. d) A 4D-printed, precursor laminate consisting of shape-shifting backbones and supportive architectures (Figure S7, Supporting Information). e) Sequential image frames of shape transformation under 90 °C heating. f) Photographs of 4D-printed face masks after transformation. Perspective view (upper left), top view (lower left), side views (lower middle and right) of the transformed face mask programmed by $f(x, y, z)$, and perspective view of the mask programmed by $f'(x, y, z) = f \times W$ (upper right). All scale bars are 3 cm.

gravity effect, into a water tank with a temperature of 90 °C (Figure 4e and Movies S3 and S4 (Supporting Information)). The 4D-printed face structures that completed the transformation showed good agreement with the models predicted in Figure 4b, in terms of the visible features such as the nose, chin, and eye lines (Figure 4f).

3. Conclusion

The programming of complex 4D architectures requires a comprehensive understanding of the target shape representation, the built-in strain tensors that induce it, and the printing technique for continuous strain programming. We have shown that the shape representation of shape-shifting 3D matters can be ultimately translated into the 4D printing parameter by a series of the F-rep-based coordination and operation, and then physically reconstructed by our newly proposed 4D printing technology. It is important to point out that the key to the function-based 4D printing is the continuous programming of the strain and printing parameters so that these two factors can be defined in function form. The proposed 4D printing technology demonstrated that the thermomechanical training and the corresponding ε_r can be continuously controlled by a single printing parameter, namely the extrusion feed rate C, without any pre- or postprocessing steps. Demonstrations ranging from functionally represented 2D curves to complex 3D surfaces, defined in function form by curve fitting and function operation, verified the feasibility of our approach. While this study focused on the continuous 4D printing of SMPs that inherently have the theoretical ε_r limit of 75%, the attainable shape configuration and function type could be further extended by exploiting materials that respond to various stimuli with high strain rates. Our findings lay the foundation for the future shape-design framework of intelligent material architectures that transform to suit targeted environments and applications.

4. Experimental Section

Materials and Apparatus Setup for 4D Printing: Polyurethane-based SMP filaments with a diameter of 1.75 mm, glass transition temperature of 55 °C, and the maximum tensile strain of 400% were obtained from SMP Technologies Inc., Tokyo, Japan. The specimens were printed based on the fused deposition modeling (FDM) method using an off-the-shelf 3D printer (Original Prusa i3 MK3S, Prusa Research) with a 0.4 mm diameter nozzle. To ensure good adhesion during printing, in previous studies, beds were generally heated to a temperature close to T_g;[6] however, this often induced relaxation of the programmed strain of deposited SMPs and, consequently, the bending of the laminate due to undesired strain mismatched. In this study, the bed was not heated; instead, a smooth polyetherimide sheet was used, which provided sufficient adhesion between the bed and the deposited materials. Thereby, specimens that exhibited 1D linear shrinkage were obtained and became a source for a more intuitive quantification of the relationship between printing parameters and recovery strain.

G-Code Modification for 4D Printing Parameter Control: Based on the understanding of the FDM 3D printing process, a scheme was established to explicitly control the printing parameters. In general, a FDM 3D printer converted the target geometry of a 3D object into an array of unit lines to print it. In this conversion process (i.e., slicing), a unit line was postulated to be a long cuboid-type fiber whose cross-section was uniformly fixed along the overall object with width w and height h (Figure S2a, Supporting Information). To print this unit line, G-code was assigned to control the extrusion and deposition process of the printer. Given w and h, the relative movement speed of the nozzle, F, length of the supplied filament (with diameter D) before extrusion, E, and length of the extruded filament (with diameter d), e, were the fundamental factors governing the 4D printing mechanism (Figure S2b, Supporting Information). It was noted that E and e were correlated by the following relation: $e = D^2/d^2E$. As opposed to F which could be independently controlled across the target object or the unit line, e was a dependent parameter that corresponded to the volume of a unit line. Specifically, an extruder motor synchronously supplied an amount of SMP equivalent to the volume of a unit line within t (the printing time for unit line) which was determined by F. Then, the following equation could be established: $4\pi d^2 e = whFt$ or $e/t = 4whF/\pi d^2$. Meanwhile, the unit line was generated by the swiping of the as-extruded fiber during deposition, at which the fiber experienced additional mechanical stretching (λ_{dep}) depending on the cross-sectional area ratio between the as-extruded and the as-deposited fibers; that is, $\lambda_d = \pi d^2/4wh$. The experimental results revealed that ε_r responding to C could be tuned by adjusting λ_d from 1 to 2.09 (Figure 2f). Importantly, the specific λ_d value uniformly applied to the overall specimen could be determined with the given w and h of the unit line, while the target geometry of the specimen remained the same. The results showed that with the same λ_d, the ε_r responses with varying w or h were identical (Figure S3, Supporting Information). As a result, C could be controlled by F with constant λ_d using the following relation

$$C = e/t = \frac{F}{\lambda_{dep}} \tag{8}$$

Given the one-to-one correspondence between ε_r and C, as described in Figure 2, a computer-aided 4D printing design tool was developed to continuously control C along the printing path, which in turn enabled spatioselective ε_r programming. The program operated with the following workflows: i) importing and arranging the G-code pregenerated in commercial slicers, ii) selecting a particular line or layer to modify; iii) assigning user-defined target strain functions based on the ε_r–C relation; iv) reconstructing and visualizing the modified G-code and saving. For the specificity of the strain function assignment, the programming method splitted up the total printing path into a set of equidistant unit segments, and assigned F values calculated from the ε_r–C relation to each corresponding G-code. Therefore, the programming resolution of ε_r tensors was determined by the length of this unit path segment, which was theoretically limited by the nozzle diameter (0.4 mm). In the experiments, this value was fixed to 1 mm to achieve reliable 4D printing.

Recovery Strain (ε_r) Analysis: A set of 4D-printed SMP specimens [30 mm (length) × 10 mm (width) × 2 mm (height)], where the extruded filaments were aligned in the longitudinal direction, were prepared to explore the programmable regime of the recovery strain (Figure S5a, Supporting Information). To provide free boundary conditions at both ends of the specimens, a handle was printed at the end of the specimen and attached to a gripper to keep the specimen in place. Shape transformation was conducted by immersing the 4D-printed specimen into a 90 °C (>T_g) water tank for 5 min which was sufficient for the saturation of the shape-shifting process. The total recovery strain was obtained by measuring the length change between the as-printed and shape-transformed samples (Figure S5c, Supporting Information). At least three repeated experiments were carried out under the same condition to verify the effectiveness of the ε_r–C relation.

Specific ε_r profiles programmed into the specimen were investigated by surface strain analysis. First, speckle patterns were randomly formed on the as-printed specimens by spray coating. The heat-induced actuation of the specimens were recorded and 25 image frames were extracted. The relative displacements of each speckle pattern of the resultant sequential image frames were analyzed by an image-processing tool (Vic-2D 2009, Correlated Solutions). The strain field data were calculated and visualized from the relative displacements based on Hencky strain tensors.

Curvature Analysis: To characterize the curvature response, a set of 4D-printed precursor laminates [100 mm (length) × 8 mm (width) × 0.8 mm (thickness)] were prepared with different built-in bilayer strain functions, as shown in Figure 3. The side of each laminate was marked black to easily track the change in arc length by image processing. To minimize the sagging effect, all samples were placed horizontally with handles attached to the gripper, which provided a free boundary condition. After actuation, the calibrated side-view images of the transformed 2D shapes were captured with a digital camera. A custom MATLAB code was used to extract the boundary nodes for the curvature calculation. From a series of parametrized biquadratic polynomials $x(t)$ and $y(t)$ which fitted the polygons for each node and adjacent nodes, the curvature was calculated using the following equation

$$\kappa = \frac{y'x'' - x'y''}{\left(x'^2 + y'^2\right)^{\frac{3}{2}}} \tag{9}$$

F-Rep 4D Printing of 3D Geometry: The F-rep 4D printing strategy for arbitrary 3D surfaces was based on parallel slicing and F-rep (Figure 4a,b). A target 3D surface was usually divided into slices with equidistant width (19 slices for a 3D face mask demonstration; see Figure 4), and the F-rep for each slice was implemented by curve fitting with the 8th-order Fourier series (Figure S8 and Table S1, Supporting Information). The target curvature function and the corresponding necessary built-in strain profiles, defined 1D along the slice, could be derived by interpretable function operations. To relate the domain of the target curvature function, $\{x\}$, with the programming domain, $\{s\}$, it was assumed that the infinitesimal length of a unit slice could be approximated by $\sqrt{dx^2 + dy^2}$ $(= dx\sqrt{1+f_{i,x}^2})$ (Figure S1, Supporting Information). Then, the relation between dx and ds could be expressed as follows

$$dx\sqrt{1+f_{i,x}^2} = ds\left(1 - \frac{\varepsilon_t + \varepsilon_b}{2}\right) \tag{10}$$

Combined with the definition of the bilayer strain function translated by the Timoshenko equation, i.e., $\varepsilon_t + \varepsilon_b = \frac{4h}{3}|\kappa| + 2\varepsilon_{min}$, the relation could be simplified as Equation (6). The results were significant in that the target 3D shape function was directly related to the built-in strain profile of its 2D precursor. Specifically, slices with a total thickness of 1.2 mm were rationally selected to minimize the sagging effect while reducing the programmable range of the curvature function to −0.3 to 0.3. Each slice consisted of six layers in total: three for the top layers (thickness of 0.6 mm) and the other three for the bottom layers (thickness of 0.6 mm).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

S.-J.A. and J.B. contributed equally to this work. S.-J.A., J.B., and K.-J.C. conceived the idea. S.-J.A. and J.B. designed and fabricated the 4D-printed precursors. S.-J.A., J.B., and H.-J.J. conducted the experiments. S.-J.A., H.-J.J., and D.-Y.L. contributed to data interpretation. J.-H.J. developed the 4D printing software. S.-J.A., J.B., and K.-J.C. wrote the paper with input from all authors.

Data Availability Statement

Research data are not shared.

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